Jiajun Wang

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PhD student in Mathematics

Institut Élie Cartan de Lorraine (IECL), Université de Lorraine, Nancy, France

jiajun.wang@univ-lorraine.fr | kazusa000.github.io | GitHub: kazusa000

Research interests

Control theory, backstepping transformations, feedback design, numerical analysis, partial differential equations, optimization, and scientific computing.

Education

Université de Lorraine Oct. 2026 – present

PhD student in Mathematics, Institut Élie Cartan de Lorraine

Université Paris-Saclay Sep. 2024 – Sep. 2026

Master studies in Mathematics and Applications: Analysis, Modeling, Simulation

Coursework: PDE analysis, finite element and boundary element methods, finite- and infinite-dimensional optimization, and CPU/GPU high-performance computing.

Jiangnan University Sep. 2020 – Jun. 2024

BSc in Information and Computational Science

Research experience

Backstepping transformations and feedback control Oct. 2026 – present

Doctoral research, IECL, Université de Lorraine

- Study constructive transformations and feedback design for finite-dimensional linear systems with single or multiple inputs.
- Prepare theoretical work as a manuscript and begin numerical comparisons, including invertible neural networks for computing transformations.

Shape optimization for hydrodynamic resistance in Stokes flow Apr. 2026 – Aug. 2026

Research internship, LS2N / CNRS, Nantes, France

- Studied shape optimization methods for Stokes flow and hydrodynamic resistance.
- Implemented numerical optimization using the Rodin finite element library.

Optimization methods for direction-of-arrival estimation Mar. 2024 – Dec. 2024

School of Science, Jiangnan University, China

- Investigated numerical optimization for high-resolution signal estimation under mixed noise.
- Contributed to model design and numerical evaluation.

Robust low-rank modeling and sparse noise removal Mar. 2023 – May 2024

School of Science, Jiangnan University, China

- Studied iterative optimization for low-rank matrix recovery and signal decomposition.
- Led numerical experiments and manuscript writing.

Parameter identification of a first-order RC equivalent circuit model Sep. 2022 – Mar. 2023

Undergraduate innovation research project, Jiangnan University, China

- Designed an identification algorithm using numerical optimization and fractional-order gradient methods.
- Performed data analysis, simulation validation, and manuscript writing.

Publications

Journal articles

1. Xie, P., Chen, J., Cheng, L., **Wang, J.** (2026). *Three-Variable Symmetric Alternating Direction Method of Multipliers for Non-Separable Convex Optimization*. Asian Journal of Control. doi:10.1002/asjc.70174.

2. Xie, P., Chen, J., **Wang, J.**, Mao, Y., Zhu, G., Yi, Y. (2026). *Alternating Direction Method of Multipliers for Direction of Arrival Estimation Under Mixed Noise*. Signal Processing, 247, 110648. doi:10.1016/j.sigpro.2026.110648.

3. **Wang, J.**, Chen, J., Zhu, Q. (2024). *The Application of the Accelerated Proximal Gradient Descent Algorithm for the Solution of the Weighted Schatten-p Norm in Sparse Noise Extraction*. Circuits, Systems, and Signal Processing, 43, 5169–5190. doi:10.1007/s00034-024-02697-z.

Manuscript

- Chen, J., **Wang, J.**, Gan, M., Xu, H., Na, J. (2026). *Spectral Partitioning for Hierarchical Identification of Ill-Conditioned Systems: A Power-Iteration-Based Approach*. Manuscript.

Conference paper

- **Wang, J.**, Cheng, L., Chen, J. (2023). *Parameter Identification of First-Order RC Equivalent Circuit Model Based on Improved Fractional-Order Gradient Descent*. 35th Chinese Control and Decision Conference (CCDC), 293–298.

Software and personal projects

- **Rodin and experiment interfaces.** Numerical research tools for finite element computation and Stokes shape optimization. Rodin; levelset-stokes-console.
- **Innerlife.** Experimental virtual-person runtime with long-term memory, emotion, relationship state, and an inspection interface. GitHub repository.

Skills and languages

Programming: Python, MATLAB, C++, C, Web development.

Methods and tools: finite elements, boundary elements, CPU/GPU computing, Lean.

Languages: Chinese (native), English (C1), French (B2), Japanese (basic).

Other interests

Violin (level 8/10), piano (beginner), and music arrangement. Self-taught arrangement and original music shared online.

Contact

jiajun.wang@univ-lorraine.fr

Additional academic contact: jiajun.wang621@gmail.com.